

Trabajo Final 2 — Algoritmos Evolutivos

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1 Resultados

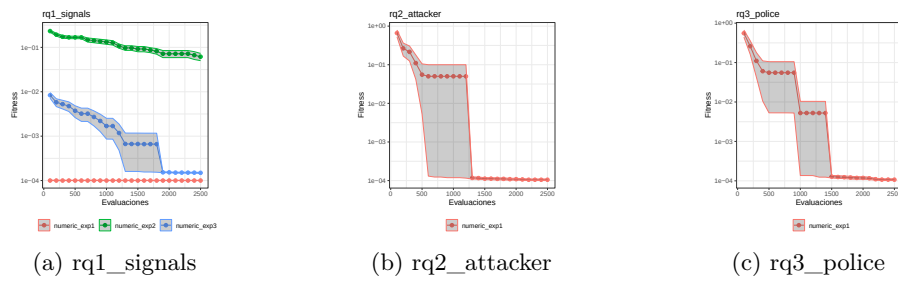


Figure 1: Evolución de fitness promediada (30 ejecuciones y 10 simulaciones por ejecución)

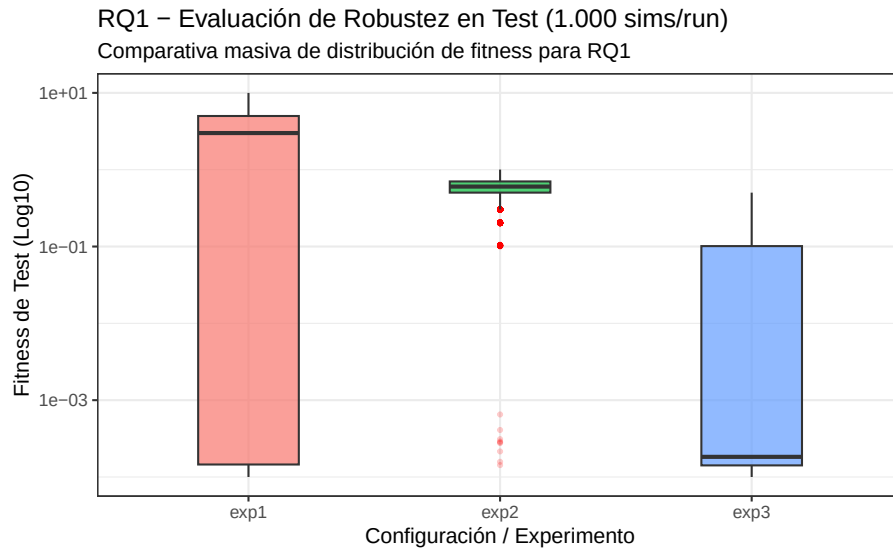


Figure 2: RQ1 (Señales): Evaluación de robustez en test de estrés.

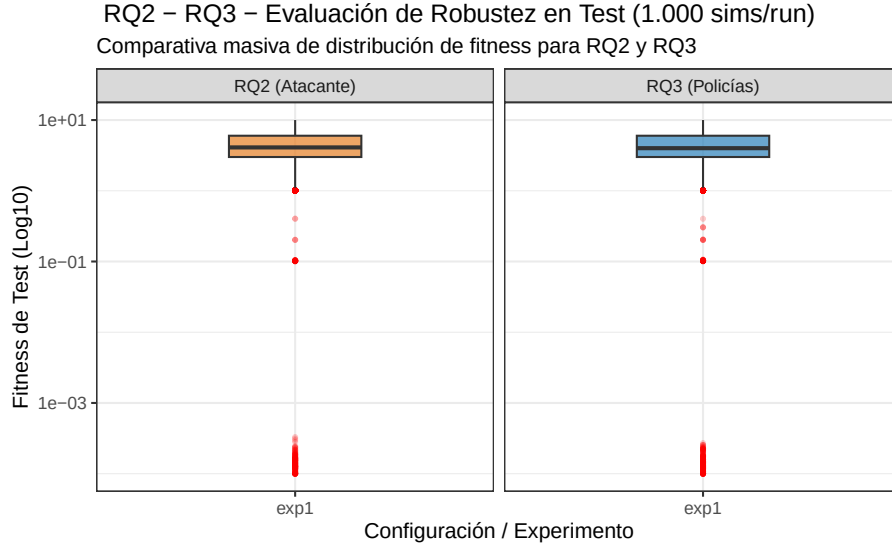


Figure 3: Evaluación de robustez en test para RQ2 (Atacante) y RQ3 (Policías) agrupando todas las ejecuciones (1.000 evaluaciones por run).

Table 1: Comparativa del Mejor Fitness en Entrenamiento frente al Test de Estrés (1.000 sims).

Instancia	Exp.	Run	Train (Fitness)	Test (Mean)	Test (Median)	Test (IQR)
RQ1 (Signals)	exp1	run_0	0.000064	0.000450	0.000142	0.000038
RQ1 (Signals)	exp1	run_1	0.000071	0.000890	0.000164	0.000042
RQ1 (Signals)	exp1	run_2	0.000067	0.000533	0.000126	0.000028
RQ1 (Signals)	exp1	run_3	0.000068	0.000430	0.000121	0.000029
RQ1 (Signals)	exp1	run_4	0.000060	0.000528	0.000116	0.000031
RQ1 (Signals)	exp1	run_5	0.000068	0.000647	0.000131	0.000031
RQ1 (Signals)	exp1	run_6	0.000059	0.000542	0.000132	0.000035
RQ1 (Signals)	exp1	run_7	0.000056	0.000533	0.000123	0.000032
RQ1 (Signals)	exp1	run_8	0.000059	0.000538	0.000126	0.000035
RQ1 (Signals)	exp1	run_9	0.000063	0.000433	0.000124	0.000033
RQ1 (Signals)	exp1	run_10	0.000075	0.000227	0.000126	0.000029
RQ1 (Signals)	exp1	run_11	0.000073	0.000228	0.000124	0.000030
RQ1 (Signals)	exp1	run_12	0.000060	0.000420	0.000113	0.000027
RQ1 (Signals)	exp1	run_13	0.000060	0.000471	0.000159	0.000043
RQ1 (Signals)	exp1	run_14	0.000060	0.000838	0.000118	0.000029
RQ1 (Signals)	exp1	run_15	0.000067	0.000687	0.000169	0.000048
RQ1 (Signals)	exp1	run_16	0.000070	0.000331	0.000126	0.000030
RQ1 (Signals)	exp1	run_17	0.000055	0.000528	0.000115	0.000026
RQ1 (Signals)	exp1	run_18	0.000068	0.000629	0.000117	0.000025
RQ1 (Signals)	exp1	run_19	0.000060	0.000634	0.000122	0.000031
RQ1 (Signals)	exp1	run_20	0.000061	0.000943	0.000126	0.000034
RQ1 (Signals)	exp1	run_21	0.000071	0.000445	0.000138	0.000039
RQ1 (Signals)	exp1	run_22	0.000056	0.000524	0.000116	0.000028
RQ1 (Signals)	exp1	run_23	0.000067	0.000229	0.000126	0.000031
RQ1 (Signals)	exp1	run_24	0.000063	0.000537	0.000127	0.000029
RQ1 (Signals)	exp1	run_25	0.000063	0.000558	0.000148	0.000035
RQ1 (Signals)	exp1	run_26	0.000059	0.000628	0.000117	0.000026
RQ1 (Signals)	exp1	run_27	0.000062	0.000228	0.000123	0.000031
RQ1 (Signals)	exp1	run_28	0.000053	0.000425	0.000115	0.000026
RQ1 (Signals)	exp1	run_29	0.000063	0.000633	0.000121	0.000028
RQ1 (Signals)	exp2	run_0	0.101799	0.557121	0.602386	0.299242
RQ1 (Signals)	exp2	run_1	0.101951	0.618298	0.603587	0.200247
RQ1 (Signals)	exp2	run_2	0.101062	0.573285	0.602735	0.200395
RQ1 (Signals)	exp2	run_3	0.102360	0.591640	0.603189	0.200221

Table 1: Comparativa del Mejor Fitness en Entrenamiento frente al Test de Estrés (1.000 sims). (*continued*)

Instancia	Exp.	Run	Train (Fitness)	Test (Mean)	Test (Median)	Test (IQR)
RQ1 (Signals)	exp2	run_4	0.104427	0.633882	0.603971	0.200383
RQ1 (Signals)	exp2	run_5	0.000189	0.610029	0.604021	0.200390
RQ1 (Signals)	exp2	run_6	0.102481	0.549745	0.602060	0.299274
RQ1 (Signals)	exp2	run_7	0.102910	0.623494	0.603158	0.200396
RQ1 (Signals)	exp2	run_8	0.000430	0.646217	0.603598	0.299249
RQ1 (Signals)	exp2	run_9	0.000321	0.578474	0.602844	0.200254
RQ1 (Signals)	exp2	run_10	0.000245	0.600482	0.602998	0.200168
RQ1 (Signals)	exp2	run_11	0.103255	0.599489	0.603291	0.200281
RQ1 (Signals)	exp2	run_12	0.103540	0.635485	0.604014	0.200603
RQ1 (Signals)	exp2	run_13	0.102066	0.637924	0.604142	0.200503
RQ1 (Signals)	exp2	run_14	0.103924	0.591856	0.603074	0.200256
RQ1 (Signals)	exp2	run_15	0.101608	0.619902	0.603360	0.200374
RQ1 (Signals)	exp2	run_16	0.000489	0.544922	0.503540	0.299070
RQ1 (Signals)	exp2	run_17	0.000431	0.581671	0.602966	0.200366
RQ1 (Signals)	exp2	run_18	0.000417	0.616679	0.603602	0.200434
RQ1 (Signals)	exp2	run_19	0.000314	0.542125	0.503269	0.200353
RQ1 (Signals)	exp2	run_20	0.102611	0.636298	0.603375	0.200220
RQ1 (Signals)	exp2	run_21	0.000412	0.605161	0.602993	0.200122
RQ1 (Signals)	exp2	run_22	0.101744	0.628113	0.603615	0.200242
RQ1 (Signals)	exp2	run_23	0.102770	0.658719	0.702981	0.200386
RQ1 (Signals)	exp2	run_24	0.101988	0.570994	0.602844	0.200635
RQ1 (Signals)	exp2	run_25	0.101966	0.557343	0.602288	0.298661
RQ1 (Signals)	exp2	run_26	0.102079	0.586031	0.603068	0.200379
RQ1 (Signals)	exp2	run_27	0.102369	0.609114	0.603300	0.200224
RQ1 (Signals)	exp2	run_28	0.201039	0.531568	0.503113	0.200594
RQ1 (Signals)	exp2	run_29	0.100964	0.621004	0.604215	0.200498
RQ1 (Signals)	exp3	run_0	0.000166	0.041844	0.000197	0.101611
RQ1 (Signals)	exp3	run_1	0.000117	0.018331	0.000141	0.000063
RQ1 (Signals)	exp3	run_2	0.000157	0.044508	0.000219	0.101953
RQ1 (Signals)	exp3	run_3	0.000137	0.035454	0.000195	0.101345
RQ1 (Signals)	exp3	run_4	0.000137	0.043937	0.000194	0.101642
RQ1 (Signals)	exp3	run_5	0.000137	0.054992	0.000231	0.102486
RQ1 (Signals)	exp3	run_6	0.000156	0.032954	0.000208	0.101001
RQ1 (Signals)	exp3	run_7	0.000133	0.027291	0.000170	0.000129
RQ1 (Signals)	exp3	run_8	0.000157	0.027271	0.000164	0.000128
RQ1 (Signals)	exp3	run_9	0.000181	0.032909	0.000198	0.100893
RQ1 (Signals)	exp3	run_10	0.000160	0.054449	0.000234	0.102677
RQ1 (Signals)	exp3	run_11	0.000157	0.033812	0.000179	0.101495
RQ1 (Signals)	exp3	run_12	0.000173	0.032298	0.000183	0.101001
RQ1 (Signals)	exp3	run_13	0.000152	0.031514	0.000173	0.100959
RQ1 (Signals)	exp3	run_14	0.000144	0.027853	0.000167	0.000132
RQ1 (Signals)	exp3	run_15	0.000150	0.030562	0.000173	0.000171
RQ1 (Signals)	exp3	run_16	0.000135	0.027512	0.000156	0.000131
RQ1 (Signals)	exp3	run_17	0.000164	0.039054	0.000212	0.101277
RQ1 (Signals)	exp3	run_18	0.000145	0.030605	0.000181	0.100891
RQ1 (Signals)	exp3	run_19	0.000126	0.036233	0.000180	0.101613
RQ1 (Signals)	exp3	run_20	0.000140	0.026400	0.000168	0.000107
RQ1 (Signals)	exp3	run_21	0.000148	0.028330	0.000172	0.000118
RQ1 (Signals)	exp3	run_22	0.000127	0.035950	0.000170	0.101464
RQ1 (Signals)	exp3	run_23	0.000148	0.033519	0.000179	0.100971
RQ1 (Signals)	exp3	run_24	0.000137	0.022800	0.000156	0.000096
RQ1 (Signals)	exp3	run_25	0.000125	0.033204	0.000194	0.100951
RQ1 (Signals)	exp3	run_26	0.000121	0.028840	0.000168	0.000144
RQ1 (Signals)	exp3	run_27	0.000152	0.026778	0.000160	0.000111
RQ1 (Signals)	exp3	run_28	0.000152	0.034556	0.000198	0.101303
RQ1 (Signals)	exp3	run_29	0.000151	0.047780	0.000220	0.102144
RQ2 (Attacker)	exp1	run_0	0.000071	4.632327	5.000101	2.897003
RQ2 (Attacker)	exp1	run_1	0.000105	4.617961	5.000074	2.999887

Table 1: Comparativa del Mejor Fitness en Entrenamiento frente al Test de Estrés (1.000 sims). (*continued*)

Instancia	Exp.	Run	Train (Fitness)	Test (Mean)	Test (Median)	Test (IQR)
RQ2 (Attacker)	exp1	run_2	0.000074	4.299343	4.000229	2.101632
RQ2 (Attacker)	exp1	run_3	0.000150	5.455497	6.000083	2.999874
RQ2 (Attacker)	exp1	run_4	0.000131	4.237520	4.000166	2.101015
RQ2 (Attacker)	exp1	run_5	0.000098	4.173171	4.000161	2.000062
RQ2 (Attacker)	exp1	run_6	0.000109	4.535902	5.000061	2.999926
RQ2 (Attacker)	exp1	run_7	0.000144	4.584477	5.000042	2.999863
RQ2 (Attacker)	exp1	run_8	0.000119	4.334839	4.000251	2.102351
RQ2 (Attacker)	exp1	run_9	0.000083	4.083118	4.000141	2.000065
RQ2 (Attacker)	exp1	run_10	0.000098	4.658685	5.000091	2.897211
RQ2 (Attacker)	exp1	run_11	0.000107	4.034705	4.000129	2.000033
RQ2 (Attacker)	exp1	run_12	0.000091	4.100629	4.000128	2.000089
RQ2 (Attacker)	exp1	run_13	0.000112	4.728049	5.000110	2.797257
RQ2 (Attacker)	exp1	run_14	0.000099	4.154161	4.000144	2.000095
RQ2 (Attacker)	exp1	run_15	0.000073	4.842491	5.000149	2.000054
RQ2 (Attacker)	exp1	run_16	0.000116	4.211427	4.000174	2.000115
RQ2 (Attacker)	exp1	run_17	0.000087	4.560876	5.000084	2.999899
RQ2 (Attacker)	exp1	run_18	0.000096	4.546847	4.153061	2.999911
RQ2 (Attacker)	exp1	run_19	0.000100	4.335981	4.000214	2.999928
RQ2 (Attacker)	exp1	run_20	0.000106	4.519703	4.102942	2.999897
RQ2 (Attacker)	exp1	run_21	0.000087	4.442360	4.101259	2.999827
RQ2 (Attacker)	exp1	run_22	0.000130	4.952765	5.000146	2.000092
RQ2 (Attacker)	exp1	run_23	0.000092	3.951057	4.000113	2.000065
RQ2 (Attacker)	exp1	run_24	0.000124	4.267950	4.000174	2.102475
RQ2 (Attacker)	exp1	run_25	0.000099	4.737913	5.000109	2.000067
RQ2 (Attacker)	exp1	run_26	0.000096	4.227298	4.000165	2.000154
RQ2 (Attacker)	exp1	run_27	0.000111	4.170900	4.000158	2.000131
RQ2 (Attacker)	exp1	run_28	0.000104	4.629842	5.000082	2.897099
RQ2 (Attacker)	exp1	run_29	0.000086	3.968390	4.000117	2.000073
RQ3 (Police)	exp1	run_0	0.000080	4.050766	4.000135	2.000091
RQ3 (Police)	exp1	run_1	0.000117	4.766558	5.000116	2.794509
RQ3 (Police)	exp1	run_2	0.000102	4.589166	5.000075	2.999825
RQ3 (Police)	exp1	run_3	0.000115	4.446263	4.101009	2.999894
RQ3 (Police)	exp1	run_4	0.000071	3.971736	4.000122	2.000034
RQ3 (Police)	exp1	run_5	0.000087	3.938608	4.000113	2.000056
RQ3 (Police)	exp1	run_6	0.000131	4.649418	5.000088	2.897879
RQ3 (Police)	exp1	run_7	0.000133	4.390242	4.102064	2.999915
RQ3 (Police)	exp1	run_8	0.000059	4.089078	4.000152	2.000059
RQ3 (Police)	exp1	run_9	0.000146	4.587645	5.000077	2.999813
RQ3 (Police)	exp1	run_10	0.000111	4.405201	4.104371	2.999923
RQ3 (Police)	exp1	run_11	0.000114	4.777329	5.000117	2.174462
RQ3 (Police)	exp1	run_12	0.000158	4.734658	5.000122	2.000057
RQ3 (Police)	exp1	run_13	0.000116	4.492733	4.104815	2.999906
RQ3 (Police)	exp1	run_14	0.000077	4.294965	4.000172	2.102673
RQ3 (Police)	exp1	run_15	0.000084	4.657595	5.000092	2.797394
RQ3 (Police)	exp1	run_16	0.000101	5.041105	5.000196	2.000101
RQ3 (Police)	exp1	run_17	0.000116	3.655562	4.000061	2.897497
RQ3 (Police)	exp1	run_18	0.000081	4.292480	4.000193	2.127520
RQ3 (Police)	exp1	run_19	0.000084	4.756950	5.000118	2.000095
RQ3 (Police)	exp1	run_20	0.000067	4.229663	4.000186	2.025218
RQ3 (Police)	exp1	run_21	0.000119	4.178398	4.000168	2.000169
RQ3 (Police)	exp1	run_22	0.000069	4.012396	4.000141	2.000054
RQ3 (Police)	exp1	run_23	0.000096	4.567873	5.000075	2.897684
RQ3 (Police)	exp1	run_24	0.000139	4.206122	4.000157	2.000146
RQ3 (Police)	exp1	run_25	0.000120	4.328968	4.100964	2.104543
RQ3 (Police)	exp1	run_26	0.000078	4.131008	4.000139	2.000079
RQ3 (Police)	exp1	run_27	0.000068	4.355241	4.000264	2.203388
RQ3 (Police)	exp1	run_28	0.000074	4.809865	5.000152	2.000129
RQ3 (Police)	exp1	run_29	0.000090	4.068613	4.000141	2.000111

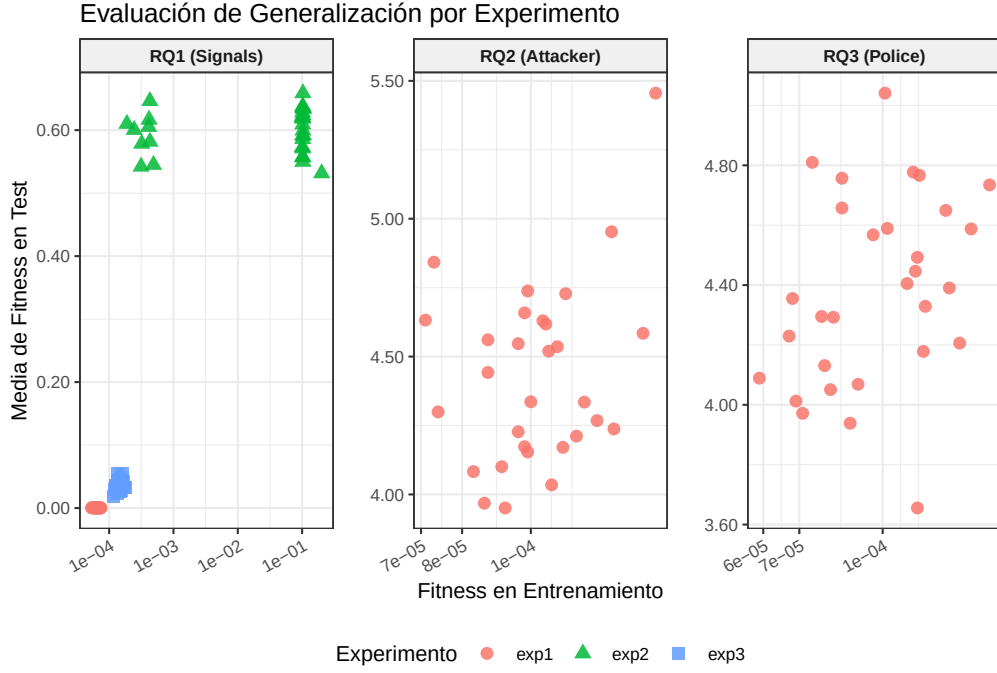


Figure 4: Correlación entre Fitness de Entrenamiento y Test por Experimento.

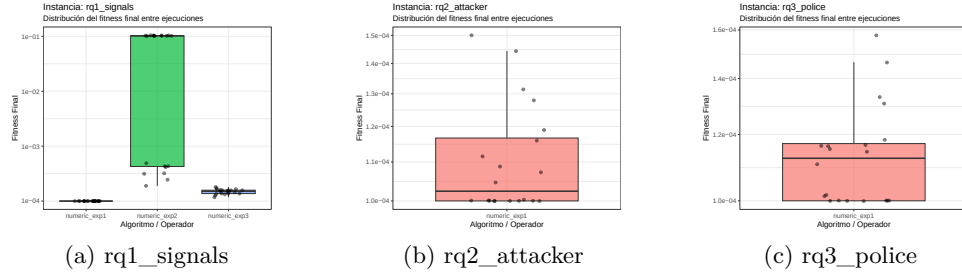


Figure 5: Comparativa de eficiencia: Distribución de fitness final (Menor es mejor)

Table 2: Resultados globales de las simulaciones. Cada columna muestra el mejor valor encontrado (x^*), la mediana ($\tilde{x}$), la media ($\bar{x}$) y el error estándar de la media ($\sigma_{\bar{x}}$). Una estrella $\star$ marca al mejor algoritmo en cada instancia, y un símbolo $\bullet$ indica significación estadística.

	numeric_exp1				numeric_exp2				numeric_exp3		
	x^*	$\tilde{x}$	$\bar{x} \pm \sigma_{\bar{x}}$		x^*	$\tilde{x}$	$\bar{x} \pm \sigma_{\bar{x}}$		x^*	$\tilde{x}$	$\bar{x} \pm \sigma_{\bar{x}}$
rq1_signals	5.51e-05	6.32e-05	6.42e-05 $\pm$ 1.28e-06	$\star$	1.89e-04	0.102	0.062 $\pm$ 0.011	$\bullet$	1.17e-04	1.51e-04	1.49e-04 $\pm$ 3.59e-06
rq2_attacker	7.05e-05	1.02e-04	1.05e-04 $\pm$ 4.94e-06	$\star$	-	-	N/A	-	-	-	N/A
rq3_police	7.08e-05	1.12e-04	1.07e-04 $\pm$ 5.39e-06	$\star$	-	-	N/A	-	-	-	N/A

Table 3: Métricas operativas de evacuación y supervivencia registradas en las simulaciones de test.

RQ	Evacuados (Media)	Fallecidos (Media)	Atrapados (Media)	Tiempo Medio (s)	Pasos Medios
RQ1 (Signals)	8.00	0.00	1.99	36.25 s	104.83
RQ2 (Attacker)	4.30	4.23	1.50	18.22 s	57.89
RQ3 (Police)	4.67	4.27	1.17	16.23 s	51.70

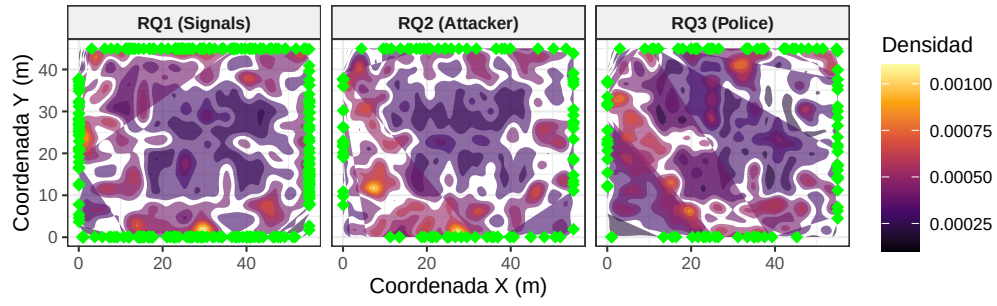


Figure 6: Mapa de calor de densidad y ocupación espacial agregada por escenario.

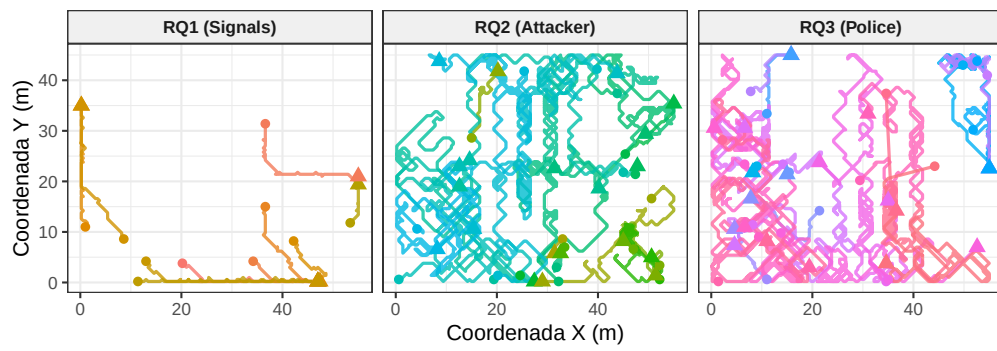


Figure 7: Trayectorias individuales de movimiento de los agentes durante la evacuación.

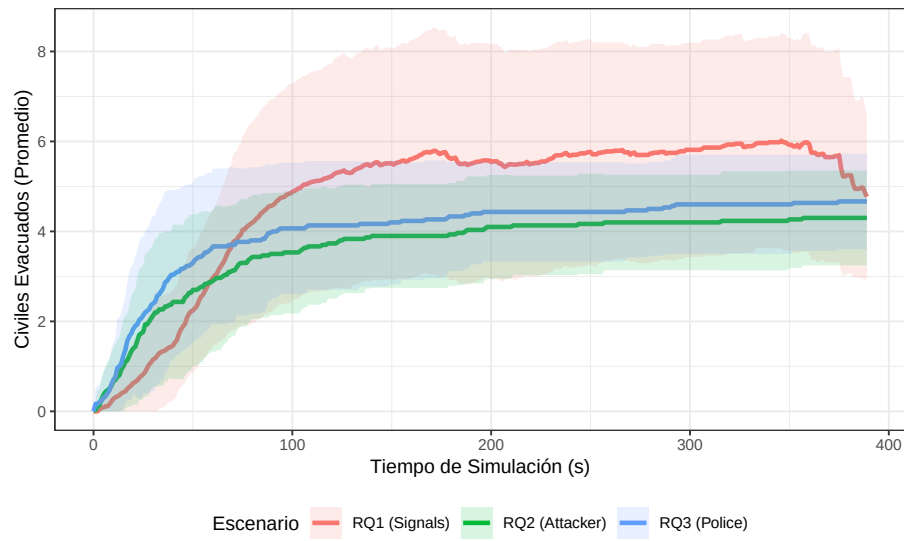


Figure 8: Tasa de evacuación acumulada promedio a lo largo del tiempo.

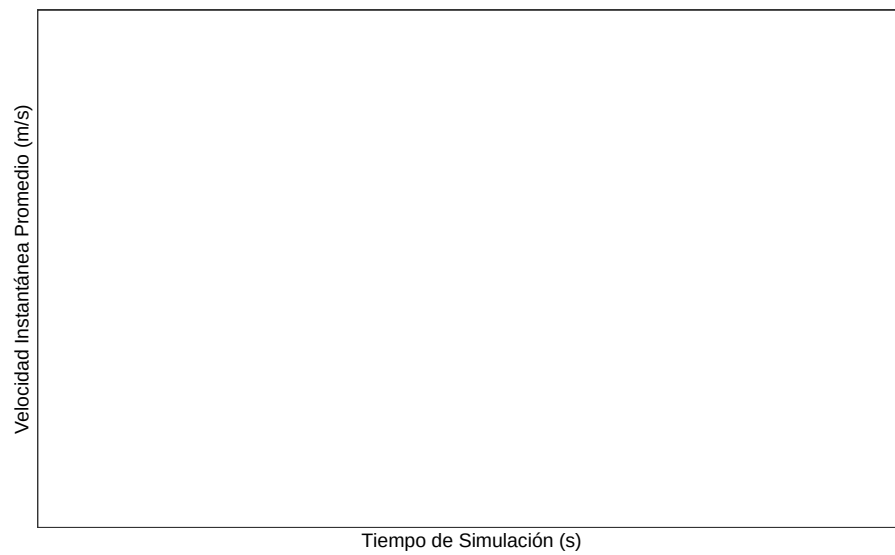


Figure 9: Evolución de la velocidad instantánea del flujo peatonal.